

Strategic Choice and Performance in Late Movers: Influence of the Top Management Team's External Ties

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ABSTRACT This work examines how the strategic choice and performance of late movers are influenced by the top management team's external ties, inside and outside the firm's industry. Using a multiyear sample of firms from the computer industry, we found that intra-industry trade-association ties of top managers led to the adoption of a resource-imitation strategy by late movers while extra-industry importation and professional-association ties led to the adoption of a resource-substitution strategy. We also found that trade association ties had a negative effect on the performance of late movers using resource-substitution, while top managers' service on other firms' boards had a positive influence on the performance of those firms. Our findings not only confirm the importance of the role of fit in strategy and performance, but they also highlight the importance of the management and control of boundary-spanning activities.

INTRODUCTION

Much of the literature on the timing of market entry has focused on first movers, which may be related to the strong reliance on theory from I/O economics and the resource-based view where barriers to entry and valuable, inimitable resources are assumed to drive sustained profitability. Consequently, relatively little attention has been paid to second or late movers (Schnaars, 1994). While there is an identifiable body of empirical work on late movers (e.g. Cho et al., 1998; Shamsie et al., 2004; Shankar et al., 1998) it remains relatively sparse.^[1] From the work that has been done, it has been shown that 'order of entry effects are not insurmountable' (Bowman and Gatignon, 1996, p. 240) and that early second movers are able to extract short-run rents from consumers (Deltas and Zacharias, 2006). When product features of late-entrant offerings are comparable to those of first movers the latecomers can have superior performance (Zhang and Markman, 1998), and

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innovative late movers can also create a 'sustainable advantage' (Shankar et al., 1998). Underpinning this success of late movers is the fact that first and late movers behave differently (Robinson and Chiang, 2002). Robinson et al. (1992) found that the resources of first and late movers tend to be different, and Shamsie et al. (2004) found that resources (e.g. firm size) and strategic positioning (e.g. product price) were key determinants of success. Consequently, for late movers, there exists specific combinations of resources and strategies (i.e. fit) that determines their success. Our work contributes to this perspective by empirically examining the effect that such a fit has on the performance of late movers.

Research has revealed that the first firm to enter a market will often achieve technological leadership, preemption of assets, and the creation of buyers' switching cost (Frynas et al., 2006; Lieberman and Montgomery, 1988, 1998). They also achieve a larger market share than late movers and are able to sustain higher prices (Kalyanaram and Gurumurthy, 1998; Kerin et al., 1992; Lilien and Yoon, 1990; Makadok, 1998; Robinson, 1988; Robinson and Fornell, 1985; Tufano, 1989). Less well publicized is the very high mortality rate of market pioneers (Golder and Tellis, 1993; Tellis and Golder, 1996) that arises from first-mover disadvantages such as the free-rider effects enjoyed by late movers, the resolution of technological and market uncertainty, potential shifts in a technology or customer needs, and first-mover organization inertia (Golder and Tellis, 1993; Kalyanaram and Gurumurthy, 1998; Lieberman and Montgomery, 1988; Lilien and Yoon, 1990; Porter, 1985; Shankar et al., 1998; Tellis and Golder, 1996). Scholars have interpreted these first-mover disadvantages as the advantages enjoyed by late movers (e.g. Kalyanaram and Gurumurthy, 1998; Lieberman and Montgomery, 1988, 1998), which late movers can utilize to adversely affect the durability of first-mover advantage by finding ways to copy the first mover at a lower cost (Porter, 1985), or by finding creative ways to market their goods (Kalyanaram and Gurumurthy, 1998; Lieberman and Montgomery, 1988, 1998), by quickly interpreting market feedback to respond to first-mover innovations (Greve, 1998). As this overview of the literature shows, most of the previous work on entry strategies has been economically oriented while little systematic research has been conducted on how managers influence the outcomes. To fully understand late-movers' strategies, it is imperative that we also consider the role played by senior management.

Upper-echelons research traditionally has been concerned with how top management team (TMT) demographic heterogeneity affects decision outcomes on things such as innovation, organization, and strategy (e.g. Bantel and Jackson, 1989; Carpenter et al., 2004; Hambrick, 2007; Hambrick et al., 1996). For example, team heterogeneity provides cognitive resources for better decision quality while hampering team-decision processes by causing more friction (Williams and O'Reilly, 1998). Beyond this work on demographics, the importance of TMT social ties is starting to be more widely recognized, particularly social ties in the form of boundary-spanning relationships; for example, Collins and Clark (2003) examined its effect on firm performance when combined with HR practices, and Geletkanycz and Hambrick (1997) examined how the ties of the TMT influence strategy selection and subsequent firm performance. However, the literature on sequential entry strategy has barely acknowledged the influence of TMT external ties despite its theoretical and practical importance (Frynas et al., 2006). This work is designed to address that issue.

Adding to and extending the research on the importance of TMT ties (e.g. Bantel and Jackson, 1989; Finkelstein and Hambrick, 1990, 1996; Hambrick and Mason, 1984; Kor, 2003), we examine the effect that external TMT ties have on strategy selection and the subsequent performance of late movers. While late movers imitate the product and service offerings of first movers, we are interested in the way in which they achieve that. Specifically, do they achieve it by deploying similar resources to the first movers (a resource-imitation strategy) or do they achieve it by substituting different resources to produce those imitative products or services (a resource-substitution strategy)? And then, having determined that, does the fit between the TMT ties and resource selection affect firm performance? We start with a review of the literature to establish our main constructs before developing theory and hypotheses, describing our methods, and reporting the results of analysis of 766 competitive responses to first-mover actions. We conclude with a discussion of our findings and the implications for research and practice.

LATE MOVERS AND TMT EXTERNAL TIES

Late-Mover Strategy and Performance

Success for firms at the business level is usually seen in terms of establishing and sustaining a competitive advantage (cf. Barney, 1991; Porter, 1985). Although first-mover advantage is implicit in the resource-based view, strategy scholars recently have suggested that the concepts of sequential entry and resource-based theory are related conceptual frameworks that can benefit from even closer linkage (e.g. Kor, 2003; Lieberman and Montgomery, 1998; Murthi et al., 1996; Robinson et al., 1992) insofar as first movers need to build barriers to imitation (Lee et al., 2000). However, such barriers can be overcome through competitive action (Reed and DeFillippi, 1990). Whereas first movers in an industry try to defend their resource base, 'challengers' (late movers) seek to 'appropriate it, through imitation, transfer and substitution of resources' (Mathews, 2002, p. 40). Mathews (2002) goes on to explain that the imitation of resources occurs through such things as being able to duplicate technology trajectories, and that substitution can occur from things like the entrepreneurial recombinations of resources. Thus, in 'competitive landscapes', there exists equifinality among resources (Peteraf and Bergen, 2003) because substitute resources can have the same strategic implications and, assuming they are no more costly to develop, then they will at least lead to competitive parity (Barney, 1991; McEvily et al., 2000).

It is through the deployment of resources that firms compete in their industries (Chandler, 1962; Finkelstein and Hambrick, 1990; Hambrick, 1980; Oster, 1982; Porter, 1980, 1985). Previous empirical research dealing with strategic choice, TMTs, or both (e.g. Finkelstein and Hambrick, 1990; Geletkanycz and Hambrick, 1997; Harrigan, 1985; Schendel and Patton, 1978) has identified the indicators of key resource deployments as advertising intensity (advertising expense/sales), capital intensity (fixed assets/number of employees), plant and equipment newness (net plant and equipment/gross plant and equipment), R&D intensity (R&D expense/sales), overhead efficiency (selling, general, and administrative expense/sales), and financial leverage (total debt/equity). Advertising intensity, capital intensity, plant and equipment newness, and R&D

intensity are all indicators of the allocation and management of firm resources across marketing, innovation, and capacity-expansion activities. Overhead efficiency captures expense structure, and financial leverage reflects an organization's approach to capital management.

We define resource imitation as the reproduction or copying of a first mover's resources in order to duplicate their advantage (Haunschild, 1993, 1994; Hunt, 2000, 2002). Examples of such imitation include Lockheed Martin's use of political influence to duplicate Arianspace's political coup of gaining access to a Russian launch location (Frynas et al., 2006). Also, from Schnaars's (1994) case analyses of late movers, Miller's and Coors' brewing skills allowed them to duplicate Rheingold's light beer, and GE had production skills that were comparable to EMI's, which meant that GE was able to imitate and eventually overcome EMI's lead in CAT scanners.

In contrast, resource substitution is defined as the use of alternative resources to achieve a given outcome (Barney, 1991, 1995; Dosi and Marengo, 1993; McEvily et al., 2000). Resource substitution by late movers can thus render obsolete the original asset stocks of a first mover (Dierickx and Cool, 1989), and is often considered to be a creative activity that has a substantial impact on the returns of both late and first movers as it reshapes the competitive landscape by allowing late movers to overtake the pioneer (Berndt et al., 1995; Yip, 1982). While resource substitution by late movers may be viewed as innovative – in Mathews' (2002) terms, the entrepreneurial recombination of resources – the innovation actually is driven by the actions of first movers (Hunt, 2000, 2002). Whereas the pioneering move of a first mover mainly results from an autogenous internal impulse for growth, resource substitution by a late mover is a responsive innovation that is directly prompted by the learning process that accompanies competition and that can result in the creation of new resources that enable it to compete in terms of relative efficiency, relative value, or both (Hunt, 2002). Canon's challenge to Xerox is an example of resource substitution. Canon attacked Xerox's advantage in photocopiers by developing the capability to manufacture high-quality copiers, which substituted for Xerox's extensive technical-service capabilities (McEvily et al., 2000; Porter and Ishikura, 1983). Resource substitution also occurred when Merrill-Lynch elected to outsource its information-technology development when trying to catch up with first mover Charles Schwab in online stock trading, and when Chevron and Mobil used their offshore oil production skills to overcome Shell-BP's land-based oil production lead in Nigeria (Mahnke et al., 2006).

TMT External Ties

Research on the roles played by CEOs and TMTs in determining organizational outcomes is a well-established stream of work (e.g. D'Aveni, 1990; Finkelstein and Hambrick, 1990, 1996; Useem, 1984; Wiersema and Bantel, 1992; see also Carpenter et al., 2004, for a summary of research in the area). In general, there is agreement that these managers set direction for the firm, choose the markets in which the firm will participate, and identify the resources it subsequently will nurture and deploy (Castanias and Helfat, 2001; Kor and Mahoney, 2000; Mahoney and Pandian, 1992). Senior managers exercise discretion when determining strategy (Finkelstein and Hambrick,

1996; Penrose, 1959; Rowe, 2001; Waller et al., 1995; Whittington, 2003) and affect the consequent outcomes from it by influencing the behaviours, thoughts, and feelings of those with whom they work (Green et al., 2003; Peters, 2001). Thus, it is these managers who make the difference between success and failure (Charan and Tichy, 1998; Tichy, 1997).

On the question of whether the CEO or the broader TMT is a better predictor of organizational outcomes, the empirical evidence suggests that the full team has a greater effect (e.g. Bantel and Jackson, 1989; Tushman et al., 1985). Accordingly, researchers have expanded their focus to include the entire upper echelon (e.g. Grimm and Smith, 1991; Kor, 2003; Wiersema and Bantel, 1992), which typically includes the officers of the corporation at senior vice-president and above (Goll et al., 2001). In keeping with these prior studies, the present work also focuses on the TMT.

The members of the TMT play an important role through their contact with external groups (Collins and Clark, 2003; Gargiulo and Benassi, 2000; Rangan, 2000; Smith et al., 1994). They create social capital that is derived from 'cooperative exchanges' in social ties, which becomes a valuable resource for the organization (Coleman, 1988). Under conditions of bounded rationality and environmental uncertainty, decision makers look to their counterparts in an effort to draw meaning from numerous and often ambiguous environmental cues (Festinger, 1954). They construct a psychological map for their own immediate context by relying on the experiences, definitions, and interpretations bestowed on similar contexts by their counterparts (Berger and Luckmann, 1967).

Research has shown that there are benefits to be gained from executives' personal and professional relationships, and which come in the form of direct influence on such things as access to resources and indirect effects such as learning and knowledge transfer. Direct effects are evident in Uzzi's (1999) work which revealed how executives' social ties can result in easier access to financing and lower interest rates on loans, and in the findings of Hillman et al. (1999), which showed how executives' service in the US Government had a positive effect on firm performance from things like reduced uncertainty, lower transaction costs, and increased amounts of legitimacy and prestige for the firm. Indirect effects – the ones we are concerned with in this work – are evident in Uzzi's (1996) work on executive ties and their implications for performance. He stated that information exchange among executives 'boosts a firms' transactional efficiency and responsiveness to the environment' and that 'social relations make information credible and interpretable, imbuing it with qualities and value beyond what is at hand' (p. 678). Testifying to the validity of this observation are findings on the importance of external linkages for the transfer of such knowledge in other settings. For example, Cho et al. (1998) explored the impact of external ties and found for a sample of Japanese and Korean late-mover semi-conductor firms that the personal linkages within *kieretsu* and *chaebols* led to knowledge sharing through social interactions and to the cross-subsidization of ideas which, of course, have implications for the formulation of firm strategy.

The boundary-spanning relationships of the TMT have thus been considered an important determinant of strategy formulation because they serve as conduits for information that shape managerial views of the environment and contribute to the set of alternatives from which strategic choices are made (Geletkanycz and Hambrick, 1997).

Further, they are reinforced through the use of common language (Pondy, 1977), shared experiences, and professional networks (Galaskiewicz, 1985) and, consequently, researchers have observed that firms tend to adopt the same practices as those organizations to which they are linked via the external ties of the TMT (e.g. Palmer et al., 1993). Together, the evidence from prior studies suggests that the external ties of top managers constitute important channels for informational influences that shape the TMT's frame of reference and, in turn, that shapes strategic choice. Specifically, the external ties of the TMT can provide firsthand insight into the need for change, as well as the approaches other firms have used to negotiate critical contingencies (Child and Smith, 1987; Hambrick et al., 1993).

THEORY DEVELOPMENT AND HYPOTHESES

TMT External Ties and the Strategic Choice of Late Movers

Granovetter (1973) observed that personal contacts may reinforce existing perspectives and insights or, conversely, may not if they expose actors to novel ideas and opportunities. Furthermore, different types of external ties – 'intra-industry ties' where top managers' boundary spanning is inside their industry and 'extra-industry ties' where their boundary spanning is outside the firm's industry – are seen as having different effects on the strategic choice of firms (Geletkanycz and Hambrick, 1997). Strategic and environmental information drawn from intra-industry sources may well bear a marked similarity to top managers' own knowledge and perceptions of the environment and opportunities within it. Hambrick (1982) noted that there exists a homogeneity in top managers' views, which arises from a commonality of knowledge of technologies in the industry, competitive conditions, and which strategies are successful (Castanias and Helfat, 2001; Kor, 2003; Schefczyk and Gerpott, 2001).

Beyond these technologies, competition, and strategies are competencies – the skill and asset interactions on which successful firms base their strategy (Hofer and Schendel, 1978). By definition, skills can be tacit in nature and lead to causal ambiguity which, in essence, creates difficulties in understanding the relationship between business inputs and outputs (cf. Reed and DeFillippi, 1990). However that ambiguity arguably will be attenuated by the common views that Spender (1996) called 'shared recipes', thus suggesting that strategies emerge as a function of managers' similar (tacit) experiences amassed through industry tenure (hence Hambrick's (1982) observation on homogeneity).

Accordingly, the ties to entities within the industry subject top managers of late movers to an abundance of information about successful practices. As a result, the tendency for shared industry views, coupled with the shared recipes and homogenizing influences that emerge in social interaction, suggests that strong intra-industry ties of the TMT of late movers are likely to lead to a preference by those top managers for resource imitation of the first mover. Therefore,

Hypothesis 1: For late-mover firms, there is a positive relationship between the TMT's intra-industry ties and the adoption of a resource-imitation strategy.

While ties inside the industry expose managers to information about successful first movers, extra-industry experience, linkages, or ties increase their ability to formulate strategies that deviate from standard industry practice. Extra-industry contacts do not rely on the same frame of reference. Rather, their views of the environment, business practices, and even goal setting are shaped by different experiences (Granovetter, 1973). In Spender's (1996) terms, they do *not* share similar recipes. While the lack of a shared recipe will make it difficult replicating what they have observed, replication would be unlikely to be useful. Instead, and per Granovetter (1973), what they have learned acts as a catalyst for adaptation and change. Consequently, extra-industry contact by top managers exposes them to new models that expand the range of strategic options, thereby allowing the TMT not only to identify and assess emerging opportunities but also to design new strategies. Therefore,

Hypothesis 2: For late-mover firms, there is a positive relationship between the TMT's extra-industry ties and the adoption of a resource-substitution strategy.

TMT External Ties and the Performance of Late Movers

Burns and Stalker's (1961) idea that fit is the key to improving performance has been a central tenet of strategic management since the seminal works of Andrews (1971) and Hofer and Schendel (1978), and particularly important within that is the notion of fit as a 'match' (Venkatraman, 1989). Consequently, researchers have found that organizations perform well to the extent that the competencies and profiles of their top managers align with, or fit, the strategies they are pursuing (e.g. Gupta and Govindarajan, 1984; Michel and Hambrick, 1992; Pfeffer, 1972; Reed and Reed, 1989). In the present study, the same rationale applies to the performance of late movers. Specifically, TMT intra-industry contacts fit with a resource-imitation strategy because they provide the knowledge and information related to the industry environment, firms competing within that environment, and the entities that affect their operation. They also provide the opportunity to glean insight into first movers' experiences with these practices (e.g. Burt, 1987; Kor, 2003).

On the other hand, extensive extra-industry ties of the TMT, which contribute to inflows of knowledge and information pertaining to other industry environments, as well as the competitive strategies employed by firms operating within them, provide insight into those practices that would otherwise only be obtained at the expense of costly experimentation or trial-and-error learning (Burt, 1987). Whereas top managers with a high level of intra-industry ties will fixate on the norms and practices of first movers in the focal industry (Finkelstein and Hambrick, 1996; Geletkanycz and Black, 2001; Hambrick et al., 1993), the viability of the firm can hinge upon top managers' ability to envision a broader range of strategic options because top managers who know of few alternatives enjoy less flexibility (Hambrick and Finkelstein, 1987); hence, their creative capacity and that of the firm is constrained.

All of that being said, factors such as changing environments, conflicting informational cues, and competing goals and expectations, together with the inherent complexity of the decision-making process, often create and intensify the cognitive limitations of strategic

decision-makers (Cyert and March, 1963; March and Simon, 1958). Top managers selectively perceive only a limited number of available cues (Simon, 1955). Additionally, in an attempt to economize on search procedures, they frequently rely on established channels and on external referents to acquire information for insight into plausible strategic alternatives (Geletkanycz and Hambrick, 1997). In line with this managerial desire to simplify the decision-making process, we adopt the view that the dominant set of TMT ties will be the primary driver of strategy selection. Therefore, in terms of the effects of fit between external ties and selected strategy, intra-industry ties combined with resource imitation should improve late-mover performance, as should extra-industry ties combined with resource substitution, but intra-industry ties and resource substitution, or extra-industry ties and resource imitation, will likely have a detrimental effect.

Hypothesis 3: For late-mover firms, a match between types of TMT ties and strategy (intra-industry ties and resource imitation, or extra-industry ties and resource substitution) will improve firm performance whereas a mismatch (extra-industry ties and resource imitation, or intra-industry ties and resource substitution) will harm performance.

METHODS

Sample and Sources of Data

The sample for this study was drawn from first and late movers in the computer-equipment industry, which is characterized by rapid technological change and a highly competitive market (Angel and Engstrom, 1995; DiDominico et al., 1996; Schoenecker and Cooper, 1998; Zantout and Chaganti, 1996), and which makes it appropriate for use in this work. Also, there is a wealth of industry- and firm-specific information (Lee et al., 2000).

Our sample frame is for the period 1985–2004. The mid-1980s marked the point at which a stream of significant new product introductions started to occur in the computer industry. The initial sample, which was comprised of the largest, publicly-traded US firms in the computer industry, was identified from the cross-reference volume of *Dun and Bradstreet's Reference Book of Corporate Management*, analysts' surveys reported in *Standard and Poor's Industry Surveys*, and *Standard and Poor's Register of Corporations, Directors and Executives*, yielding an initial sample of 95 firms. Firms included in the sample had to match the four-digit SIC code for the computer industry. The SIC coding system is periodically updated to reflect structural and technological changes in the economy, and in the 1987 SIC revision the computer industry was changed to Electronic Computers (SIC 3571). The firms under this new heading had previously been classified into 'Electronic Computing Equipment' (SIC 3573). Confirmation that firms were indeed computer-equipment manufacturers was done by checking *Hoover's Handbook of American Business*, and *10-K SEC Filings*.

Information for the TMT, including external ties, was drawn from *Dun and Bradstreet's Reference Book of Corporate Management*, *Standard and Poor's Register of Corporations, Directors and Executives*, as well as *10-K and Proxy Statements*. Per the norm in TMT research, we included

all individuals with titles of senior vice president and above (e.g. Carpenter and Fredrickson, 2001; Finkelstein and Hambrick, 1996; Goll et al., 2001). The average team size was 6.1. We measured TMT members' external ties at time t (i.e. the point of a new product introduction). Resource profiles and financial data for first and late movers were established using data from COMPUSTAT. In the process of merging the TMT's external ties and the companies' resource-profile information, the sample was limited to firm-pairs (first and late movers) that had data available in COMPUSTAT. Because firms went public in different years, and not all of them continued to operate through 2004, the sample did not include an equal number of firms for each year.

Measures

Identifying first and late movers. Similar to studies on market-entry strategies, we defined the first movers in the computer industry as the firms that introduced a radically new product or service that exploited a technological discontinuity (e.g. Lee et al., 2000; Lieberman and Montgomery, 1988; Tellis and Golder, 1996; Zantout and Chaganti, 1996). The first introduction of a radically new product or service is a special case of significant and pioneering innovation by first movers (Lee et al., 2000) that has been identified as a main source of first-mover advantages in many empirical as well as theoretical studies in market-entry strategy (e.g. Lee et al., 2000, 2003; Lieberman and Montgomery, 1988; Nelson and Winter, 1982; Porter, 1985; Yip, 1982).

Using LexisNexis we first identified all the introductions of 'pioneering new-product or service categories' based on 'significant innovation' for each year of the study period (e.g. Ferrier and Lee, 2002; Lee et al., 2003; Zantout and Chaganti, 1996). There were 32 new-product or service innovations by nine first movers over the 17-year time span from 1985 to 2001. In turn, those that responded to these first movers' actions were identified as late movers, yielding a total of 766 actions by 55 late movers. The selection of first and late movers was confirmed from articles that appeared in the trade or business press, and from industry monographs that detailed the history of the industry and that also identified new products as innovative, pioneering introductions (see Appendix). Again, this approach is consistent with studies on sequential entry strategies (e.g. Kor, 2003; Zantout and Chaganti, 1996).

Measuring late-mover strategies. The strategies of late movers, resource imitation and resource substitution, are on a continuum that reflects the extent to which they deviate from the resource profile of the industry's first mover. The use of a continuum for exploring resource profiles is well established in the literature (e.g. Chen, 1996; Collis, 1991; Finkelstein and Hambrick, 1990; Geletkanycz and Hambrick, 1997), and we follow this precedent. While these strategies are conceptually and philosophically separate, there exist sound theoretical reasons why the associated resource profiles should not be considered as categorical. As already noted, the resource-based view and sequential-entry strategies are conceptually related (e.g. Kor, 2003; Lieberman and Montgomery, 1998; Murthi et al., 1996; Robinson et al., 1992) and, per the existence of resource heterogeneity that is central to the resource-based view (e.g. Barney, 1991; Wernerfelt, 1984), a categorical approach to the classification of resources would be misleading

because some dissimilarities in resources among firms will always exist. We therefore employ the continuum concept whereby the resource profiles of late movers that are more similar to first movers are considered to be more imitative while those that are more different are considered to be more substitutive. Thus, a comparison of the resource profiles of the matched first and late movers at time $t + 1$ was used to measure late-mover strategy.

Per our earlier explanation of how resources have been classified in extant research, we use advertising intensity, capital intensity, plant and equipment newness, and R&D intensity as indicators of the allocation and management of firm resources across marketing, innovation, and capacity-expansion activities, and overhead efficiency and financial leverage to capture expense structure and management's approach to capital management. In combination, these ratios provide an overview of firms' competitive profiles (Finkelstein and Hambrick, 1990). Accordingly, we assessed the resource profiles of first and late movers for each year of the study period and calculated the absolute difference of each late mover's scores from each first mover's score after standardizing. Then, following Finkelstein and Hambrick (1990) who applied a similar approach to measure strategic conformity, we summed all differences from the six measures to create a single, composite indicator of strategy. Thus, and as already noted, a resource profile closer to that of a first mover (i.e. a lower summed-difference score) indicates a strategy based mostly on resource imitation while a resource profile that is substantially different (a higher summed-difference score) indicates a greater reliance on resource substitution.

By using a lagged design, delays occurring between the time when top managers' participating in external interactions that affect their strategic thinking, and then engaging in decision-making activities which are manifested in observable indicators, are accounted for (Geletkanycz and Hambrick, 1997; Haveman, 1992). In other words, the external ties of the late mover TMT were identified at time t and the relative resource profile of a late mover at time $t + 1$. While a longer time lag can be used (e.g. a two-year lag), top managers are most likely to recall and use information derived from recent interactions (Virany et al., 1992). Additionally, not only is each of the resource dimensions used in this study amenable to change in a relatively short time, but also, in a dynamic, competitive industry, like computer equipment, it is unlikely that late movers could afford to wait two years.

Performance. Performance was measured using return on assets (ROA), which is seen as useful for assessing the performance implications of business strategies because it captures the degree to which management has effectively deployed firm assets (Oster, 1990). In addition, we also performed a second set of analyses using return on sales (ROS). Although the use of accounting-based data for performance measurement has been criticized (Fisher and MacGowan, 1983; Salamon, 1985), previous studies have shown accounting-based performance to correlate with market-based performance (e.g. Kothari, 2001; Penman, 1991).

Because it is widely acknowledged that extraneous factors can introduce variability into single-year performance measures, we used two-year averages (e.g. Meyer and Gupta, 1994; Oster, 1990) that were lagged by one additional year from decisions on strategy; i.e. we captured performance measures at time $t + 2$ and $t + 3$ in recognition of

the delay between the time when strategic actions are undertaken (at time $t + 1$) and the TMT's ties at time t . Because of this lagged design, performance was measured up to and including 2004, but the time frame for TMT ties was curtailed at 2001.

External ties. To measure intra-industry ties, we counted the total number of TMT members' (1) intra-industry importations (i.e. hiring top managers from inside the focal industry) and (2) trade association ties (i.e. trade organization leadership positions held by the focal firm's top managers) at time t . Imported top managers bring with them knowledge gained through personal experience with other firms' policies and practices (Granovetter, 1988). Research suggests, however, that over time their attention becomes increasingly inward-focused (Katz, 1982). Consequently, prior empirical studies limited TMT intra-industry importation to the previous 10 years (e.g. Finkelstein and Hambrick, 1990). They also adjusted this measure using a recency factor: $(10 - \text{firm tenure at the focal firms})/10$. For example, a top manager who had been with the focal firm for seven years received an importation score of 0.3. We too adjusted for recency. In addition, as another predictor of intra-industry ties, we counted the number of trade-association leadership positions held by top managers at time t . The underpinning rationale for this measure is that trade associations constitute a forum for the establishment of industry rules of behaviour, which are influenced by the first mover (DiMaggio and Powell, 1983; Herman, 1981).

The literature on the external ties of the TMT (e.g. Geletkanycz and Hambrick, 1997; Granovetter, 1973) has identified an array of linkages that facilitate the transfer of information across firms operating in different industries including: (1) the extra-industry importation of top managers into the top management team of the focal firm; (2) top managers serving on other firms' boards outside the focal industry; (3) outside directors from outside the focal industry; and (4) memberships in professional business associations. We use these linkages as evidence of inter-industry ties.

We measured extra-industry importation at time t in the same way as intra-industry importation, including an adjustment for recency. The prevailing view of interlocking directorates sees them as vehicles for scanning the business environment (Useem, 1984) and gaining firsthand insight into other organizations' activities (Burt, 1983). In particular, top managers serving on other firms' boards directly influence the strategic decision and performance of the focal firm (Haunschild, 1993). Also, we counted all TMT members' directorships in other firms outside the focal industry at time t . Consistent with the findings of previous studies (e.g. Geletkanycz and Hambrick, 1997; Zajac, 1988), our data revealed that top managers in the sample firms had a very small number of intra-industry sent-directorate ties. Accordingly, we excluded these intra-industry sent interlocks, ensuring that the measure reflects only extra-industry sent interlocks. Although outside directors are not responsible for creating strategy, they act as advisers (Lorsch and MacIver, 1989; Mace, 1986), they bring a unique, outside perspective, and they help transfer organizational practices across industry boundaries (e.g. Davis, 1991; Haunschild, 1994; Palmer et al., 1995). Outside directors thus constitute 'received' interlocks, which we measured by counting all outside directors with primary responsibilities outside of the focal firm's industry at time t . As in the measurement of extra-industry sent interlocks, we excluded (the very few) within-industry received interlocks.

Finally, memberships in organizations that draw top managers from diverse industries are primary mechanisms for the exchange of unique information among organization leaders (e.g. Aldrich and Pfeffer, 1976; DiMaggio and Powell, 1983). Active participation in professional associations has thus been considered an important means by which to gain new ideas and perspectives. Such affiliations allow top managers to interact openly with others working in a similar professional capacity (Scott, 1985). To measure professional association ties, we counted all top managers' memberships in industry-spanning associations, such as the Conference Board and the Business Roundtable, at time t . Because the number of external ties covaries with TMT size (Geletkanycz and Hambrick, 1997), we normalized the data by dividing the number of ties by team size.

Control variables. Prior research (e.g. Bamberger and Fiegenbaum, 1996; Fiegenbaum et al., 1996; Jackson and Dutton, 1988) has demonstrated empirically that firms that achieved superior past performance are more likely to adopt less ambitious goals and more conservative strategies. In contrast, firms that had poor performance were likely to adopt more ambitious goals and aggressive strategies. Thus, we controlled for past performance.

We also controlled for diversification. In diversified firms, managers may have access to the same ideas from their colleagues for which managers in undiversified firms would need extra-industry contacts. We used a Rumeltian-like approach to measuring diversification: firms with 70 per cent or more of revenues from a single four-digit industry were classified as single/dominant, whereas those with less than 70 per cent of revenues in a single four-digit industry were classified as diversified (Rumelt, 1974).

Because TMT experience is important for strategic choices and consequent firm performance (e.g. Marlin et al., 2004), we controlled for TMT tenure using the mean number of years that members of the TMT had spent in the firm. In addition, because firms often have different resources and perform differently at different stages of development – start-up versus mature – we controlled for firm age, measured as the number of years since the firm was founded (Kor, 2003; Markman and Gartner, 2002). Firm size, measured as the log of total assets, was included as a control because it has been identified in various studies as a factor that impacts strategic choice (e.g. Dean et al., 1998; Kor, 2003). Because of the lagged nature of the dependent variables, measures at time t were introduced as additional controls. Such controls are necessary when datasets are slow to change over time, or when they may be influenced by unobserved variables (Geletkanycz and Hambrick, 1997; Judge et al., 1982). Current-year controls of lagged variables reduce the likelihood of model misspecification (known to generate biased and inconsistent estimates), and facilitate better assessment of the effects attributable to the independent variables studied. They also are important in clarifying the temporal order of measures and thus rule out the potential problem of reverse causality (Eisenmann, 2002; Sanders and Boivie, 2004; Wiklund and Shepherd, 2003). Consequently, our analysis estimates the effects of the independent variables on the dependent variables, given the prior level of dependent variables.

Analysis

The unit of analysis was the firm. The hypotheses presented in this study were tested using multiple regression. Given that interaction effects only are meaningful if the

interaction terms give a significant contribution over and above the direct effects of the independent variables, Hypothesis 3, which posits the interaction effects of the TMT's external ties and selected strategy on performance, was tested using a two-step moderated multiple-regression design, as recommended in the literature (e.g. Morgan and Piercy, 1998; Richey et al., 2004; Stone and Hollenbeck, 1989; Wiklund and Shepherd, 2003). The data were corrected for autocorrelation by applying the Cochrane–Orchutt transformation (Kmenta, 1986), and the Durbin–Watson statistic (2.02) was examined to verify that autocorrelation was not a problem (Neter et al., 1985). Each of the variables was mean-centred prior to forming the multiplicative term to alleviate any multicollinearity (Aiken and West, 1991; Cronbach, 1987). The Variance Inflation Factor also was examined to verify that multicollinearity was not a problem. Finally, plots of residuals were examined to confirm normality.

RESULTS

The means, standard deviation, and correlations for the study's key variables are reported in Table I. Table II reports the multiple-regression analysis results for the strategic choices of late movers at time $t + 1$.

The results generally provide support for both Hypotheses 1 and 2. Although intra-industry importation was not significantly related to late movers' choice of a resource-imitation strategy, we found, consistent with expectations, that intra-industry ties via trade associations were negatively related to the adoption of a resource-substitution strategy ($\beta = -0.068$, $p < 0.05$), which supports Hypothesis 1. Consistent with Hypothesis 2, hiring top managers from outside the focal industry was positively associated with the adoption of a resource-substitution strategy ($\beta = 0.164$, $p < 0.001$), as was professional-association ties ($\beta = 0.067$, $p < 0.05$). The size of extra-industry received interlocks and top managers' outside-directorship networks (those formed through their service on other firms' boards), however, did not appear to affect the tendency of late movers to choose resource substitution over resource imitation.

Several control variables showed significant effects on the strategic choice of late movers. A significant negative relationship ($p < 0.01$) between firm age and resource substitution was observed. It thus appeared that older late movers tended to be characterized by strategic profiles that remain similar to those of the successful first movers in their industry. We also found a significant negative relationship ($p < 0.001$) between firm size and resource substitution by late movers – the larger the late mover, the less likely it was to use a resource-substitution strategy. Past performance at time t exhibited a negative relationship with resource substitution at time $t + 1$ ($p < 0.01$), and thus implies that late movers that achieved high levels of performance are more likely to adopt a resource-imitation strategy rather than a more ambitious resource-substitution strategy. Diversification was not significantly related to the strategic choice of late movers. Strategy at time t was found to be positively and significantly related to the strategy at time $t + 1$ ($p < 0.001$), and it thus appears that the strategic choices of late movers were characterized by a form of inertia – what they have done before is what they tend to do in the future. That relationship was, however, affected by TMT tenure, which was positively related to the use of resource substitution by late movers, which suggests that

Table I. Descriptive statistics and Pearson correlations^a

<i>Variable</i>	<i>Mean</i>	<i>S.D.</i>	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>	<i>6</i>	<i>7</i>	<i>8</i>	<i>9</i>	<i>10</i>	<i>11</i>	<i>12</i>	<i>13</i>
1. Strategy (<i>t</i> + 1)	5.74	2.29													
2. Average ROA (<i>t</i> + 2, <i>t</i> + 3)	-0.78	12.32	-0.06												
3. Intra-industry importation	0.44	0.66	-0.03	0.02											
4. Trade association ties	0.20	0.50	-0.14	-0.04	-0.02										
5. Extra-industry importation	0.42	0.58	0.15	-0.02	0.01	-0.07									
6. TMT outside directorship	0.93	1.49	-0.01	-0.03	-0.03	0.01	0.06								
7. Outside director	1.70	1.60	-0.14	0.15	-0.18	0.21	0.05	0.15							
8. Professional association ties	0.28	0.64	0.09	0.02	-0.10	0.03	0.22	0.26	0.27						
9. TMT tenure	11.16	5.92	-0.19	0.04	-0.40	0.17	-0.28	0.29	0.32	0.08					
10. Firm age	32.50	26.47	-0.32	0.08	-0.18	0.10	-0.01	0.20	0.47	0.12	0.37				
11. Firm size	7.22	1.76	-0.50	0.11	-0.09	0.12	-0.11	0.24	0.34	-0.03	0.52	0.49			
12. Strategy (<i>t</i>)	6.12	2.40	0.57	0.06	-0.01	-0.11	0.05	0.06	-0.20	0.02	-0.11	-0.28	-0.37		
13. Performance (<i>t</i>)	0.51	20.14	-0.22	0.19	0.06	-0.03	-0.02	0.04	0.08	-0.01	0.07	0.06	0.30	-0.16	
14. Diversification	0.15	0.35	-0.15	0.09	-0.19	0.12	-0.02	0.16	0.37	0.14	0.24	0.43	0.26	-0.12	0.07

Notes: ^a Correlations greater than 0.07 are significant at the 0.05 level; N = 766.

Table II. Effects of the TMT's external ties on strategic choice^a

<i>Variables</i>	
Intra-industry importation	0.007 (0.22)
Trade association ties	-0.068 (-2.51)*
Extra-industry importation	0.164 (5.16)***
TMT outside directorships	0.043 (1.48)
Outside directors on firm's board	0.058 (1.92)
Professional association ties	0.067 (2.17)*
TMT firm tenure	0.081 (2.12)*
Firm age	-0.109 (-3.15)**
Firm size	-0.267 (-7.16)***
Strategy (<i>t</i>)	0.401 (13.48)***
Performance (<i>t</i>)	-0.073 (-2.62)**
Diversification	-0.027 (-0.86)
<i>R</i> ²	0.471
<i>F</i> -statistic	55.89***

Notes:

^a Standardized regression coefficients are reported. T-test results are in parentheses.

* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

tenure helped ameliorate any inertia. It is worth noting that this finding challenges earlier research that showed a positive relationship between short organizational tenure and strategic planning openness (e.g. Bantel, 1994; Finkelstein and Hambrick, 1990; Wiersema and Bantel, 1992). The finding in this study is not alone, however, in as much as firm tenure is a component measure of industry tenure (years spent in the firms are also years spent in the industry). In fact, industry-specific managerial experience that involves knowledge of the opportunities, threats, competitive conditions, and regulations, as well as goodwill with suppliers and customers, enhances top managers' abilities for formulating a creative strategy (Kor, 2003; Schefczyk and Gerpott, 2001). Thus a conflict between the TMT's tenure and strategic choices may occur when the TMT possesses high levels of firm-specific and industry-specific experience simultaneously. Despite the fact that managers with high levels of firm-specific experience can be entrenched by industry norms and practices (Geletkanycz and Black, 2001; Hambrick et al., 1993), the potential value that industry-specific managerial experience adds to the ability of the TMT to formulate a creative strategy (Castanias and Helfat, 2001; Cooper et al., 1994) is important. This noted, what may be of greater interest is the indication that the relationship between top managers' tenure and their strategic choices is more complex than generally believed.

Table III reports the multiple-regression analysis results for the effects of TMT ties (at time *t*) on two-year average firm performance (at time *t* + 2 and *t* + 3). Hypothesis 3 posited there were two-way interactions in such a way that late movers performed better when the type of TMT ties matched the selected strategy than when it did not. Given that strategy was measured according to resource similarity with the first mover (i.e.

Table III. Effects of the TMT's external ties on performance (N = 766)^a

	<i>Restricted model</i>	<i>Full model</i>
Intra-industry importation	0.067 (1.83)	0.075 (2.04)*
Trade association ties	-0.113 (-3.22)***	-0.151 (-4.23)***
Extra-industry importation	0.051 (1.32)	0.032 (0.77)
TMT outside directorships	-0.111 (-2.95)**	-0.102 (-2.66)**
Outside directors on firm's board	0.143 (3.66)***	0.14 (3.51)***
Professional association ties	-0.05 (-1.25)	-0.044 (-1.04)
Firm age	0.015 (0.34)	0.004 (0.09)
Firm size	0.116 (2.57)**	0.132 (2.93)**
Strategy ($t + 1$)	0.155 (4.06)***	0.13 (3.31)***
Performance (t)	0.172 (4.76)***	0.171 (4.77)***
Diversification	0.065 (1.63)	0.046 (1.14)
Interaction of strategy with ^b		
Intra-industry importation		0.013 (0.36)
Trade association ties		-0.162 (-4.40)***
Extra-industry importation		0.045 (1.03)
TMT outside directorships		0.101 (2.66)**
Outside directors on firm's board		-0.031 (-0.86)
Professional association ties		0.029 (0.69)
R^2	0.105	0.137
F -statistic	8.021***	6.993***
ΔR^2		0.032
Change in F		4.676***

Notes:^a Standardized regression coefficients are reported. T-test results are in parentheses.^b Interaction terms were added to restricted model.* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

higher scores mean more resource substitution), the expected sign for the interaction terms of intra-industry ties and strategy is negative (i.e. intra-industry ties at t leads to *worse* firm performance at time $t + 2$ and $t + 3$ when the late movers choose resource substitution over resource imitation at time $t + 1$) while that for resource substitution is positive (i.e. extra-industry ties at t leads to *better* firm performance at time $t + 2$ and $t + 3$ when the late movers choose resource substitution over resource imitation at time $t + 1$). The results show that fit between type of TMT external ties and late-mover strategies made a significant contribution over and above the restricted model ($\Delta R^2 = 0.032$, $p < 0.001$). We found a significant, negative interaction between strategy and the TMT's intra-industry trade association leadership ties ($\beta = -0.162$, $p < 0.001$).

Figure 1 is a graphic representation of the pattern of the two-way interaction using one standard deviation above and below the mean of the interacting variable to establish end points (Aiken and West, 1991). It demonstrates that the pattern of the two-way interaction was as hypothesized. In addition, we also found a significant, positive interaction between strategy and extra-industry sent interlocks ($\beta = 0.101$, $p < 0.01$). As shown in Figure 2, and as expected, the positive link between a resource-substitution strategy and

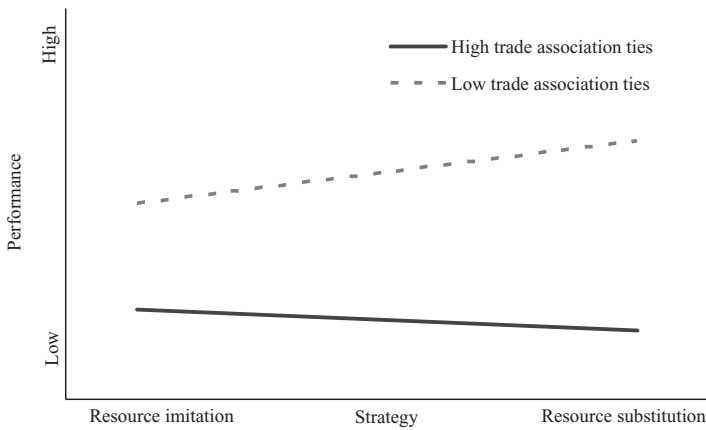


Figure 1. Trade association ties–strategy interaction for performance

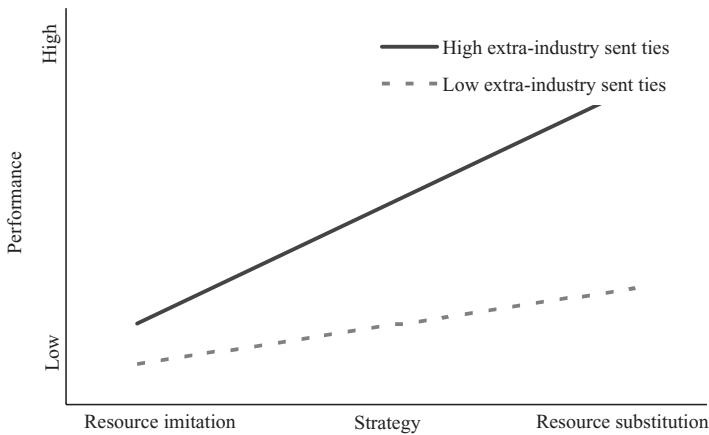


Figure 2. Extra-industry sent ties–strategy interaction for performance

the performance of late movers was stronger when top managers have high levels of extra-industry sent interlocks. However, other relationships were not supported.

In a separate analysis we used two-year averages of ROS as the measure of performance, and results were unchanged.

DISCUSSION

We sought to extend understanding of the factors that influence the strategic choices of late movers and consequent firm performance and, using a multiyear sample of first and late movers in the computer industry, we found that the external ties of the TMT contribute to the shaping of organizational strategy. Specifically, we found that intra-industry trade-association ties promoted the use of a resource-imitation strategy. Our results also show that two types of extra-industry ties, extra-industry importation and

professional association ties, were strongly related to the adoption of a resource-substitution strategy by late movers. These findings are consistent with prior observations that the hiring of top managers from outside the industry (e.g. Hambrick et al., 1993) and participation in professional associations (e.g. Palmer et al., 1995) are important for gaining divergent insights and perspectives.

However, contrary to prediction, hiring top managers from firms in the same industry did not lead to greater resource imitation. One possible explanation for this finding may be that they are all sufficiently like-minded that the importation of new members from firms operating in the same industry has little incremental impact on the strategic choice of firms. Also, the size of extra-industry received interlocks and top managers' outside-directorship networks formed through their service on other firms' board did not appear to affect the tendency of late movers to choose resource substitution over resource imitation. Specifically, the services of outside board members as well as TMT service on other firms' boards do not appear to affect strategy choices by late movers.

This study contributes to and extends the sequential-entry strategy literature by linking it with the TMT literature. Especially, the present study constitutes a first attempt at empirically examining the impact of the fit between late-movers' strategy and TMT's external ties. One form of TMT intra-industry tie examined in this study – trade association ties – was observed to have a significant, negative effect on the performance of late movers implementing a resource-substitution strategy showing that, consistent with expectations on fit, strong external ties that do not align with firm strategy will be harmful to performance. Additionally, top managers' service on other firms' boards was observed to be beneficial to the performance of late movers implementing a resource-substitution strategy, despite its significant, negative main effect on firm performance. As we argued above, the effectiveness of the TMT's external ties and their impact on firm performance should be understood and explained in terms of their fit with the strategy the firm is pursuing.

The other types of TMT external ties examined in this study – intra-industry importation, outside director, extra-industry importation, and professional association ties – were observed not to significantly affect performance when interacting with strategy. One possible explanation for this might be the different levels of utility of the different types of tie. Firms may ascribe differing values and applications (e.g. attracting critical resources, gaining organizational legitimacy) to different types of tie. Executives may well seek external ties for informational purposes for dealing with rapid changes in technology, demand, and competition. For example, several scholars have described outside directors as relatively passive participants in the strategic management of firms (e.g. Herman, 1981; Mace, 1986) – while they bring to the firm unique perspectives and insights, other advantages may be limited.

Clearly, our work parallels that of Geletkanycz and Hambrick (1997) insofar as, like them, we examined the boundary-spanning activities of executives and how that related to chosen strategy and subsequent performance. We adopted many of their approaches to the operationalization of variables and, perhaps not surprisingly, like them, we found somewhat mixed but generally supportive results for a link between TMT ties and adopted strategy. But, where they found that 'conformity' produced superior performance in the 'more uncertain computer industry' than 'deviant' or 'non-conformist'

strategies, we found that fit between the ties of the TMT and the adopted strategy was the crucial determinant of performance for late movers. In other words, resource imitation and substitution worked equally well provided there was a fit with TMT ties and, by implication, the expertise and information they brought to strategy selection. This difference between our results and the findings of Geletkanycz and Hambrick (1997) could be due to our focus on late-mover strategy and performance (Geletkanycz and Hambrick likely had first movers in their sample), the slightly different and longer time period of our study (1985–2004 vs. 1983–87) or the larger number of observations (766 responses by late movers vs. 275 ‘firm-year observations’). It should be noted that our findings do not invalidate the work of Geletkanycz and Hambrick; instead, they should be viewed as confirming and further refining the findings of their work. In particular, by addressing the contingent match between the strategies of late movers and the external ties of their top managers, this study adds evidence to a growing stream of research that demonstrates that TMT links to the external environment are among the many factors that help to shape the strategies and performance of firms.

Limitations

Limitations of the current study means that several new research questions for future investigation are raised. Firstly, given that this work used secondary data, we were unable to address individuals’ motivations for forming external ties. The broad implicit assumption of this study was that executives formed such ties to improve strategy fit and performance, but other evidence suggests that numerous motivations underlie the formation of managers’ external ties. For example, boundary spanning relationships have been linked to the management of critical resource dependencies necessary for maintaining ongoing operations (e.g. Burt, 1980; Pfeffer and Salancik, 1978; Westphal et al., 2006), interorganizational power (Pfeffer, 1987), the attainment of organizational legitimacy (Galaskiewicz, 1985), organizational learning (Levitt and March, 1988), and the enhancement of organizational prestige (D’Aveni, 1990). Future studies need to attend to these different motivations underlying network formation and delve further into the array of consequences for late-mover strategy formulation and performance.

Secondly, we were unable directly to investigate the finer details of social networks, such as frequency and/or strength of interaction in the social ties. Given that our study provided mixed support for some hypotheses (i.e. only some intra- and extra-industry ties had significant relationships with the choice of resource imitation and resource substitution), the nature of social networks might determine the effects of social ties on strategic choices (e.g. McDonald and Westphal, 2003). Future studies need to investigate further this issue to clarify the relationship between TMT social ties and late movers’ strategic choice.

Thirdly, this study and its findings are limited to large firms. The motivation for focusing upon large organizations was data availability through regularly published, archival sources and their responsiveness to the significant pioneering moves of successful first-movers. However, research suggests that the TMT’s external ties may have a greater impact on smaller late-movers (e.g. Starr and MacMillan, 1990). Inasmuch as they lack the store of resources maintained by larger firms, smaller late-movers are particularly

dependent upon top managers' social ties for their survival and growth (Higgins and Gulati, 2006). On the other hand, the work of Burgelman (1983a, 1983b, 1994) suggests that the strategic process in large, complex organizations is multi-layered – middle-level managers initiate an autonomous strategic change and determine the strategic context, whereas top managers play a crucial role of evaluation of the means and ends of strategy. Thus, study of the impact of top managers' external ties on firm-level strategic outcomes among smaller late-movers and/or their fit with the characteristics of middle level managers among larger late-movers may reveal even more insight into the relevance of boundary-spanning ties.

Fourthly, this study makes a simple distinction between first movers and all others within an industry because, as was noted earlier, it was considered a suitable approach for the purpose of the study. However, empirical findings on market-entry strategies (e.g. Boulding and Christen, 2003; Lee et al., 2003) reveal that some fast second movers can achieve a better performance than late movers (and better than some first movers). Complicating the issue even further are the arguments in the strategic-group literature (e.g. Fiegenbaum and Thomas, 1995; Greve, 1998; Hunt, 1972; Nair and Kotha, 2001) whereby membership in a particular group defines the essential characteristics of the firm's strategy and, thus, heterogeneity in performance within an industry (Peteraf and Shanley, 1997; Reger and Huff, 1993).

Finally, this study did not control for TMT demographic heterogeneity. Previous studies have shown that team heterogeneity could provide more cognitive resources from a wider range of perspectives and backgrounds (Wiersema and Bantel, 1992), which might also have influenced the strategic choices of our sample firms; i.e. the more heterogeneous the TMT of later movers, the more likely they would be to choose resource substitution than resource imitation. Indeed, Hambrick et al. (1996) suggested that demographically diverse TMTs are more likely to make more innovative decisions, but less likely to quickly respond to competitors' initiatives. It is possible that controlling for team heterogeneity may reveal a clearer picture of the relationship between the TMT's external ties and strategic choices and performance. Future studies should examine which sources of the TMT's cognitive repertoire have a significant impact on their decision-making processes and its outcomes.

Implications for Research and for Practice

As noted earlier, insufficient attention has been paid to the development and evaluation of late-mover strategies. One reason for this shortcoming may have been the lack of an objective measure for late-mover strategies. This study helps overcome this problem. The measure developed for and employed in this work, which draws on established theoretical arguments (e.g. Kor, 2003; Lieberman and Montgomery, 1998; Murthi et al., 1996; Robinson et al., 1992), can be used across industries and is thus generalizable across different environments. Our findings also provide insights for managers. For example, the results indicating that the external ties of top managers affect organizational outcomes of late movers points to the importance of considering networks of personal relations when evaluating managerial candidates. Recruitment and selection should thus be driven as much by strategic need as by the managerial skills of the applicants.

Continuing that logic, once the TMT is in place, late movers that seek to maximize performance can select either a strategy of resource imitation or resource substitution according to the basket of external ties. Or, for a given strategy, senior managers can be encouraged to cultivate the ties that fit, so that performance can be enhanced.

Conclusion

Our study makes contributions to the literature and practice by theoretically identifying and testing the linkages between TMT external ties, late movers' strategic choice, and the equifinality created by resource fit. The findings suggest that late movers with TMTs with intra-industry ties favour a resource-imitation strategy, whereas TMTs with extra-industry ties favour a resource-substitution strategy. In addition, the results imply that late movers are more likely to succeed if there is the fit between TMT external ties and the selected strategy. Thus, by highlighting the importance of the management and control of boundary-spanning activities of TMTs, this study provides insights about how late movers make their strategy decisions and when those decisions result in a favourable outcome.

NOTE

- [1] It should be noted that there also exists a significant body of work in economics on second movers (e.g. Amir and Stepanova, 2006; Gal-Or, 1985; Hoppe, 2000; Liu, 2005; Pepall, 1997; Rhee, 2006; Shinkai, 2000; Weimann et al., 2000). The primary focus of that stream of research has been mathematical modelling of second-mover entry, rather than empirical testing of entry and performance.

APPENDIX: TECHNOLOGY AND PRODUCT INNOVATION IN THE COMPUTER INDUSTRY FROM 1985 TO 2001

<i>Year</i>	<i>First move</i>	<i>Company</i>
1985	3090 Processor	IBM
	Wireless LAN technology	Apple
1986	130 MB HD	Compaq
	3.5/1.2 MB	Compaq
	First 80386-based PC compatible computer	Compaq
	First personal computer to provide Embedded SCSI	Apple
	First commercially-available 32 bit RISC-based computer	IBM
1987	3.5/1.44 MB	HP
1988	First graphics supercomputer	Apollo
	First computer using erasable optical disks	IBM
1989	SPARCstation, a low-end RISC workstation	Sun Microsystems
	First 80486-based computer	IBM
	First personal computer product using 4 megabit memory chips	IBM
	First EISA PC	Compaq
1990	64 MB RAM	HP
	Fault-tolerant VAX computer	Digital Equipment
	First 486 portable personal computer	IBM
1991	RISC-based 9000 Series 700 workstations	HP
	First 64-megabit memory chip	IBM

APPENDIX: *Continued*

1992	OS/2 2.0 operating system	IBM
	PowerPC 601 processor	IBM and Motorola
1993	First packaged speech recognition product	IBM
1994	First notebook computer with a CD-ROM drive	IBM
1995	First 12.1-inch thin-film transistor display on a laptop	IBM
1996	Wireless mouse and keyboard	Gateway
	16.1-inch colour active matrix flat-panel LCD screen	IBM
1997	15.1-inch active-matrix flat-panel display	Compaq
	First Net PC	Compaq
	First notebook computer with DVD-ROM drive	IBM
1998	275 MHz PowerPC 750 processor	IBM
	Solaris 7.0 operating system	Sun Microsystems
1999	AirPort wireless connection, based on the IEEE 802.11b standard	Apple
	Ultrastor 72X hard drive, with world record 73 GB capacity	IBM
2000	First computer featured with 1.2 GHz Athlon processor	Micron Technology
	First personal computers with standard dual processors	Apple
2001	Pen-based personal computer	IBM
	First DVD+RW drive available for personal computers	HP

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